

MAE323 Aerospace Structures

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Design of a Semi-Span Wing for Firefighting Applications

Team 6

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1 Executive Summary (EI)

This project presents the design, analysis, and fabrication of a semi-span wing for a small tilt-rotor UAV designed for firefighting applications. The aircraft is designed to carry a fire-extinguishing payload while operating in hazardous environments where human access may be unsafe. Our objective was to design a lightweight wing structure capable of withstanding the critical cruise and takeoff loading conditions while satisfying the aerodynamic and structural requirements of the mission.

The design process began by sizing the wing based on the given aerodynamic constraints. With a total aircraft mass of 1.5 kg, a cruise velocity of 15 m/s, and a maximum wing loading requirement of 8 kg/m^2 , we selected a full span of $b = 0.95 \text{ m}$, chord of $c = 0.25 \text{ m}$, planform area of $S = 0.2375 \text{ m}^2$, which gave an aspect ratio of $AR = 3.8$. Using the Clark-Y airfoil and finite-wing lift correction, the final lift coefficient was calculated as $C_l = 0.45$, with a required angle of attack of approximately 3.195° . These values were used as the basis for the rest of the structural design.

Two loading cases were considered during the design process: steady-state cruise and take-off. The takeoff condition was determined to be the most critical case since it creates a concentrated tip load on the wing. The takeoff load was calculated as $T = 9.6075 \text{ N}$, and after applying a safety factor of 1.5, the design tip load was 14.411 N. This produced a maximum root bending moment of $M_{\max} = 6.845 \text{ N} \cdot \text{m}$, which was used to guide the spar, rib, and stringer design.

The preliminary structural design used a balsa wood I-beam spar, four ribs, and six square stringers. CAD-derived section properties for the spar were used in the buckling and stress analyses, with $I_{xx} = 0.037 \text{ in}^4$, $I_{yy} = 0.015 \text{ in}^4$, and cross-sectional area $A = 0.306 \text{ in}^2$. While Spar column buckling initially suggested a rib spacing of 0.2375m, the stringer buckling analysis was more restrictive, giving a final rib spacing of 0.1583 m and four ribs total across the semi-span.

The design was then verified through direct stress, shear stress, failure analysis, and FEA. The maximum direct stress in the spar was calculated as 5.0812 MPa, and the maximum shear stress was 183.4 kPa. Applying the Coulomb-Mohr failure criterion gave a value of 0.727, which fell within the acceptable design range for the project. The FEA showed that the highest stress concentration will occur around the root rib attachment holes, indicating that the root rib and fastener region were the most critical locations for failure.

The final prototype was fabricated using balsa wood, epoxy, Monokote covering, and mechanical fasteners. The completed prototype had a semi-span of approximately 0.475 m, chord of 0.25 m, and a measured mass of approximately 83 g. During testing, the wing failed at an applied mass of approximately 1.3 kg, corresponding to a load of about 12.75 N. This was slightly below the target design load of 14.411 N, or about 88.5 percent of the expected load capacity. The failure occurred near the root attachment region, which was consistent with the FEA results showing the largest stress concentration around the root rib screw holes. Overall, the final design meets the main aerodynamic and structural goals of the project while remaining practical to manufacture/test.

2 Problem Statement(YB)

Modern urban firefighting increasingly demands access to environments that are too dangerous for human crews. When a building is engulfed in flames or structurally compromised, firefighters face significant risks, including collapse, extreme temperatures, and limited visibility due to smoke. Before, human crews would have been the only option, but now small-scale unmanned aerial vehicles (UAVs), can be a great alternative, as they can operate in structurally compromised buildings and in smoke-filled conditions without putting any first responders' lives at risk. A good solution for these types of cases is a tilt-rotor VTOL configuration, as it not only offers high forward flight efficiency but also vertical takeoff capability necessary for rapid deployment.

As the structural engineering team, our role is to ensure that the UAV's airframe can survive all of the loads and conditions that it will be exposed to throughout the mission. All of the choices made throughout the design process will determine if the aircraft can survive these conditions. Some of the characteristics that we have to consider include the wing's weight, its span, aspect ratio, etc. Weight is important, as a weight that is too heavy will diminish flight performance, while one that is too light will fail under the distributed lifting loads that will be experienced during vertical takeoff. Furthermore, the wing geometry is one of the most consequential decisions. If we choose to have a wing with a higher aspect ratio, then we can produce more efficient lift, but we are more susceptible to bending. On the other hand, a lower aspect ratio will be stiffer but will require more wing area to generate the same lift, which means increased weight for our wing. Furthermore, we also have to make decisions on the internal structure, such as the placement, number, and sizing of the wing components, including the spars, stringers, and ribs, to carry the shear, bending, and torsion that affect the wing.

The two critical load cases that will be driving our design are a uniform distributed lift load during cruise at 15 m/s with full payload, and a tip-concentrated thrust load during takeoff at 3 m/s² acceleration. Each requirement imposes a distinct bending, shear, and torsional stress distribution on the wing structure that we will be considering throughout our design process.

3 Wing Conceptual Design (YB) (TH)

3.1 First Design Phase (YB)

In this phase of our design we first established the size and geometry of our wing. The aircraft has an empty weight of 1.0 kg and carries a 500 g payload, giving a total mass of $m = 1.5$ kg and a total weight of $W = mg = 1.5(9.81) = 14.72$ N. For steady level cruise, we assume $L = W$.

One of the constraints given was a maximum wing loading of $W_s \leq 8$ kg/m², where $W_s = \text{total mass}/S$. Isolating for the area gives the minimum area constraint of:

$$S \geq \frac{m}{W_{s,max}} = \frac{1.5}{8} = 0.1875 \text{ m}^2 \quad (1)$$

Using 0.1875 as an initial estimate, we solved for the lift coefficient using $L = \frac{1}{2}\rho V^2 SC_L$ with $L = W$:

$$C_L = \frac{2W}{\rho V^2 S} = \frac{2(14.715)}{(1.225)(15)^2(0.19)} \approx 0.56 \quad (2)$$

To find the angle of attack corresponding to this C_L , we used the finite-wing lift curve relation $C_L = C_{L\alpha}\alpha + C_{L0}$, where the finite-wing lift curve slope is:

$$C_{L\alpha} = \frac{C_{l\alpha}}{1 + \frac{C_{l\alpha}}{\pi e \cdot AR}} \quad (3)$$

To evaluate this, we first calculated minimum the aspect ratio using an approximate $S = 0.19$ m² as found earlier:

$$AR = \frac{b^2}{S} = \frac{0.95^2}{0.19} = 4.75 \quad (4)$$

Substituting into the slope equation with $e = 0.7$:

$$C_{L\alpha} = \frac{2\pi}{1 + \frac{2\pi}{\pi(0.7)(4.75)}} \approx 4.11 \text{ rad}^{-1} \quad (5)$$

The zero-lift coefficient $C_{L0} \approx 0.25$ was estimated from a C_l vs. α chart of the Clark-Y airfoil. Since we now have both $C_{L\alpha}$ and C_{L0} we can solve for α :

$$\alpha = \frac{C_L - C_{L0}}{C_{L\alpha}} = \frac{0.56 - 0.25}{4.11} \approx 0.0754 \text{ rad} \approx 4.32 \quad (6)$$

With an initial estimate of α established, we then determined the final wing dimensions by graphing the two constraints in the b - c plane: the span constraint $b \leq 1$ m and the wing loading

boundary $c = 0.1875/b$ as shown in Figure 20 below.

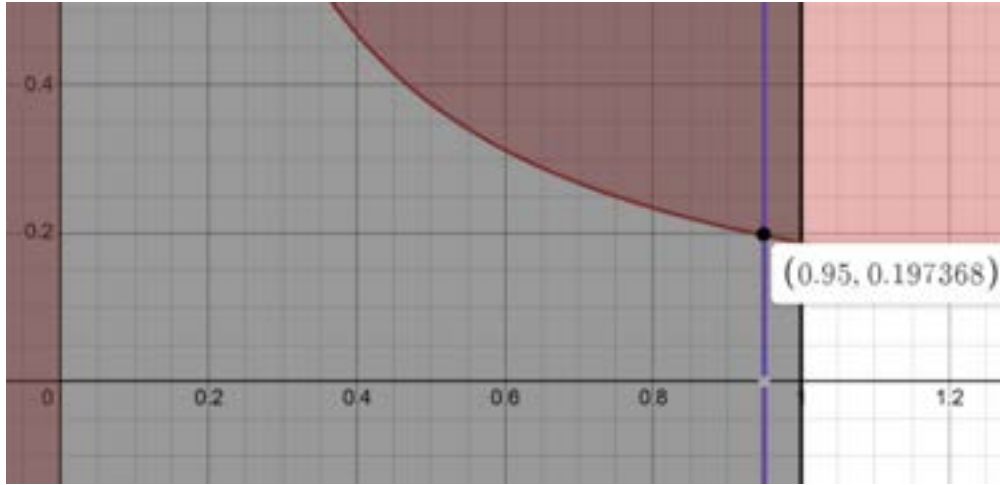


Figure 1: Graphing of the Constraints

Based on that figure we choose a span of $b = 0.95$ m since we were trying to approach the 1 m limit as close as possible to try and maximize aerodynamic efficiency. At this span, the constraint curve gives a minimum chord of $c_{min} = 0.1875/0.95 \approx 0.197$ m however we opted for a $c = 0.25$ m chord. Furthermore, with our chosen chord and span values, we are able to recalculate our lift coefficient and angle of attack to produce the desired lift using $e = 0.7$ and $C_{L_0} = 0.25$. The required lift coefficient at trim is:

$$C_L = \frac{2W}{\rho V^2 S} = \frac{2(1.5)(9.81)}{(1.225)(15^2)(0.25)(0.95)} = 0.450 \quad (7)$$

Using the linear lift model $C_L = C_{L\alpha} \alpha + C_{L_0}$, the finite-wing lift-curve slope is:

$$C_{L\alpha} = \frac{2\pi}{1 + \frac{2\pi}{\pi \cdot e \cdot AR}} = \frac{2\pi}{1 + \frac{2}{0.7 \cdot \left(\frac{0.95}{0.25}\right)}} \quad (8)$$

Solving for angle of attack:

$$\alpha = \frac{C_L - C_{L_0}}{C_{L\alpha}} = 3.195^\circ \quad (9)$$

This was done in order to provide a good manufacturing margin, and all aerodynamic values were subsequently recalculated with the updated parameters, as summarized in Table 1.

Table 1: Final wing dimensions and recalculated aerodynamic parameters

Parameter	Symbol	Value
Full span	b	0.95 m
Chord	c	0.25 m
Aspect ratio	AR	3.8
Planform area	S	0.2375 m ²
Lift coefficient	C_L	0.45
Angle of attack	α	3.195°

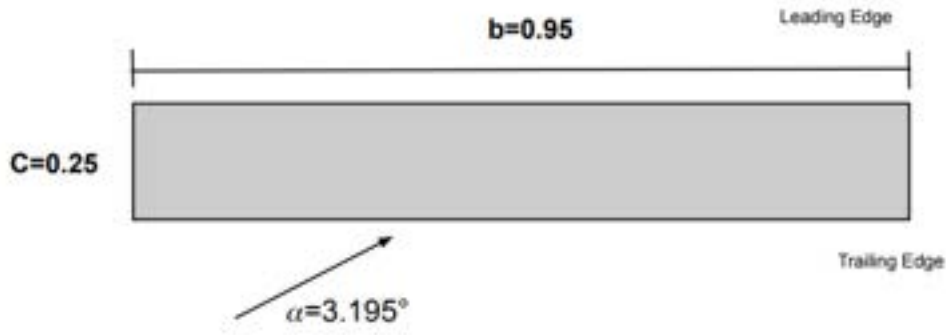


Figure 2: Visualization of the wing dimensions

3.2 Loading Values (TH)

Based on our mission summary our wing will experience two types of loads one at cruising and one at takeoff. During cruise, the aircraft is not accelerating, as we are informed that the weight of the plane is balanced by the lift ($L=W$), so $F_{net} = 0$. During takeoff, we are informed that the wing must withstand an upward acceleration of 3 m/s; therefore, F_{net} is not equal to 0. The forces acting on the wing during takeoff are the takeoff force at each wing tip and the weight of the wing. Therefore, because the wing accelerates during takeoff, while, on the other hand, experiences negligible acceleration during cruise, we conclude that the takeoff condition imparts the most significant loading on the wing, and is therefore the critical flight condition to use to size the wing's key components.

The takeoff load is calculated as follows using $g = 9.81$ m/s, $a = 3$ m/s and $m = 1.5$ kg:

$$\begin{aligned}
 F_{net} &= ma \\
 2T - W &= ma \\
 T &= \frac{m(a + g)}{2} = 9.6075 \text{ N}
 \end{aligned} \tag{10}$$

3.3 Diagrams (YB)

In the diagrams below (excluding the two overall loading diagrams), we neglect the weight of the wing, which would be a downward distributed force. We were instructed by course staff to do this in order to produce more conservative estimates of the maximum moment, but also because, in the setup of the final demonstration's critical load test, the direction of the weight of the wing is equal to that of the critical load. As a result, the diagrams below, along with the takeoff shear and moment diagrams, neglect the weight of the wing.

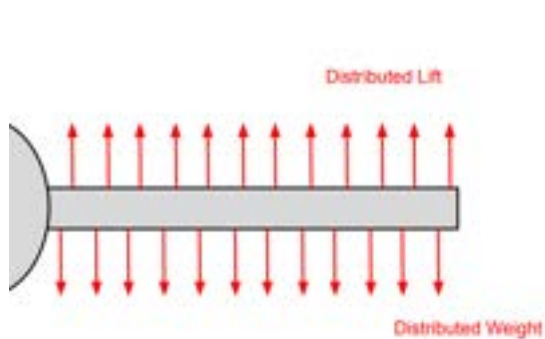


Figure 3: Cruise Loading Diagram

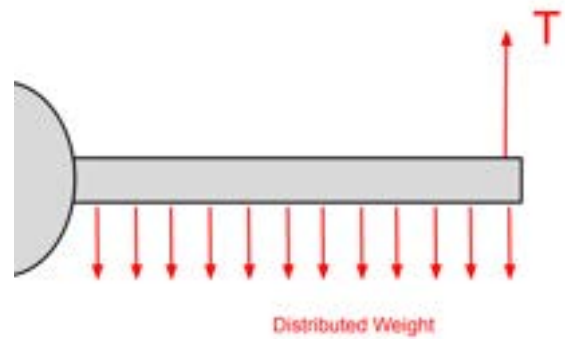


Figure 4: Takeoff Loading Diagram

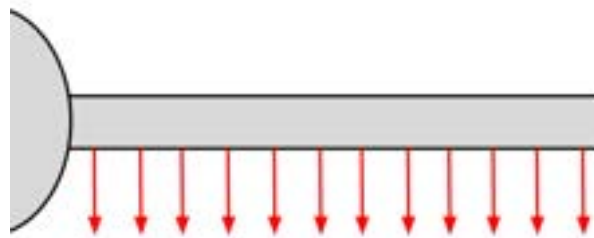


Figure 5: Distributed Cruising Load

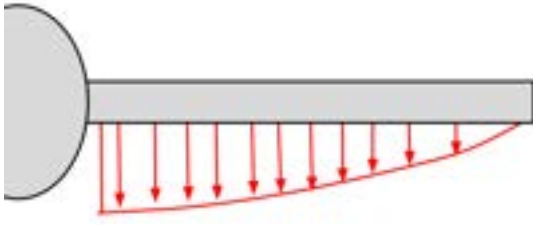


Figure 6: Cruising Moment Diagram

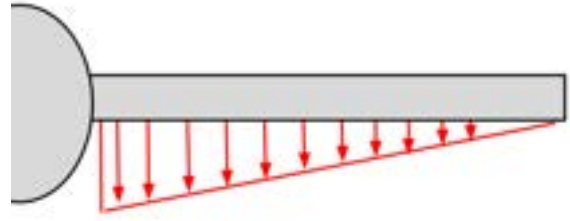


Figure 7: Cruising Shear Diagram

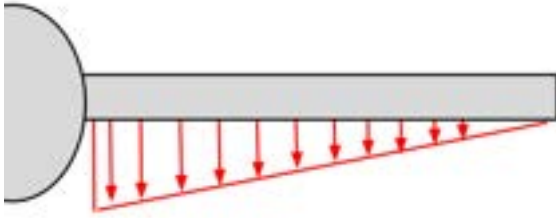


Figure 8: Takeoff Moment Diagram

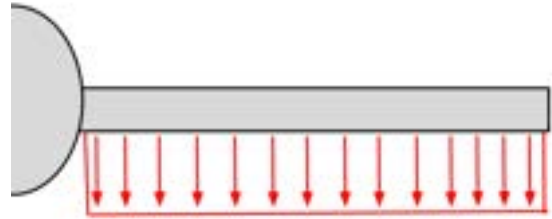


Figure 9: Takeoff Shear Diagram

3.4 Takeoff Shear and Moment Equations (TH)

As explained previously, we are neglecting the weight of the wing, so the only load considered during takeoff is the takeoff force at the wing tip T :

$$V(x) = -T \quad (11)$$

therefore:

$$M(x) = \int_x^{b/2} V(x) dx = \int_x^{b/2} -T dx = -T x \Big|_x^{b/2} = -T \left(\frac{b}{2} - x \right) \quad (12)$$

$$M(x) = -T \left(\frac{b}{2} - x \right) \quad (13)$$

The takeoff moment equation is used later in this report, in Section 4, in order to determine the maximum moment the wing experiences during takeoff. In these calculations, we apply a safety factor of 1.5 to the takeoff load, T .

4 Wing Preliminary Design (TH)

In this section, we focus on the design of our wing box components along the span and the chord, specifically the spar, ribs, and stringers. In Section 3, we determined that vertical takeoff is the most critical flight condition, as the wing experiences larger loads and moments than in the cruise condition; therefore, the analysis below is carried out only for the takeoff condition. The cross section of our wing box regarding the wing's span and stringers is constant across the span, so we analyze the respective cross section right next to the root rib, as that cross section experiences the largest moment, as shown in Section 3. This cross section is displayed in Figure 10 below:

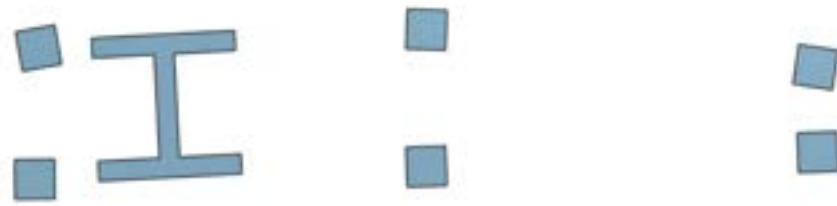


Figure 10: Spar and Stringer Cross Section of the Wing Box next to Root Rib

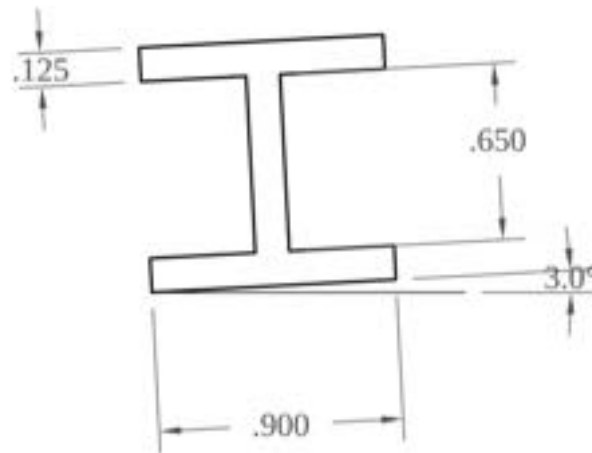


Figure 11: Cross Section of the Wing Spar

The dimensions of our I-beam cross-section are shown above: the I-beam has a total height and length of 0.9 in and a thickness of 0.125 in (1/8 in). Therefore, during the construction phase, we used 0.9 in flanges and a 0.65 in web. These were the maximum dimensions that we could reasonably fit within the boundaries of the airfoil shape. A thickness of 0.25 in would have shortened the web to 0.4 in, making it less than half the height of the I-beam. Additionally, we designed the I-beam to be tilted 3° with respect to the chord line. Without this tilt, the bottom-right corner and top-left corner of the I-beam would be positioned too close to the edge of the airfoil boundary,

concentrating stress at these points and causing premature failure. For the stringers, we used 0.25 in x 0.25 in rods as they provided significantly more support than the 0.125 in rods. We also used a square cross section for the stringers to maximize the support they were able to provide. After designing the I-beam above in OnShape, we used the CAD software to find the values below:

$$I_{xx} = 0.037 \text{ in}^4 \quad I_{yy} = 0.015 \text{ in}^4 \quad I_{xy} = I_{yx} = 0.001 \text{ in}^4 \quad A = 0.306 \text{ in}^2$$

I_{xy} and I_{yx} are not equal to 0 due to the slight tilt in our I-beam; however, because these values are so small, they can be assumed to be negligible in the following analysis. The unit conversions from inches to meters used in calculations below can be found in Appendix Section A.1

4.1 Buckling Analysis

4.1.1 Part 1: Spar Column Buckling and Rib Spacing

The critical stress for column buckling is given by:

$$\sigma_{cr} = \frac{\pi^2 E}{\left(\frac{l_e}{r}\right)^2} \Rightarrow \sigma_{cr} = \frac{\pi^2 E I_{\min}}{l_e^2 \cdot A} \quad (14)$$

where $r = \sqrt{I/A}$.

The bending moment distribution along the span is:

$$M(x) = -T \left(\frac{b}{2} - x \right) \quad (15)$$

The maximum moment at the root ($x = 0$) is:

$$M_{\max}(x = 0) = -T \cdot \frac{b}{2} = -1.5(9.6075) \cdot \frac{0.95}{2} = -6.845 \text{ N}\cdot\text{m} \quad (16)$$

using a suggested safety factor of 1.5 applied to the takeoff load, T.

Bending of the wing caused by takeoff is about the x -axis, so the maximum bending moment on the wing is:

$$\sigma_z = \frac{My}{I_{xx}} \Rightarrow \sigma_{cr} = \frac{|M_{\max}| y_{\max}}{I_{xx}} = 5.0812 \text{ MPa} \quad (17)$$

For the column buckling of our spar, we assumed pinned-pinned boundary conditions, since the wing will have a rib on each end, and we assume the ribs prevent transverse displacement of the spar but offer no rotational restraint. Therefore, $l_e/l = 1$, and using the column buckling formula with $I_{\min} = I_{yy}$, $E = 3.4 \text{ GPa}$ from MatWeb material properties for American balsa wood, and

the unit conversions in Appendix Section A.1:

$$l_e = \left[\frac{\pi^2 E I_{yy}}{\sigma_{cr} \cdot A} \right]^{1/2} = 0.457 \text{ m} = l \quad (18)$$

Using this critical buckling length, the number of ribs along the half-span is:

$$N_{\text{ribs}} = \frac{b/2}{l} = \frac{0.475}{0.457} = 1.04 \Rightarrow N_{\text{ribs}} = 2 \Rightarrow \text{Spacing} = \frac{b/2}{2} = 0.2375 \text{ m} \quad (19)$$

Rounding up from the calculated number of ribs to two ribs gives $N_{\text{ribs}} = 3$, since one rib is needed at each end of the wing; therefore, our rib configuration will consist of a rib on each end of the wing and a rib in the middle, yielding a rib spacing of 0.2375 m.

4.1.2 Part 2: Local Plate Buckling of Web and Flange

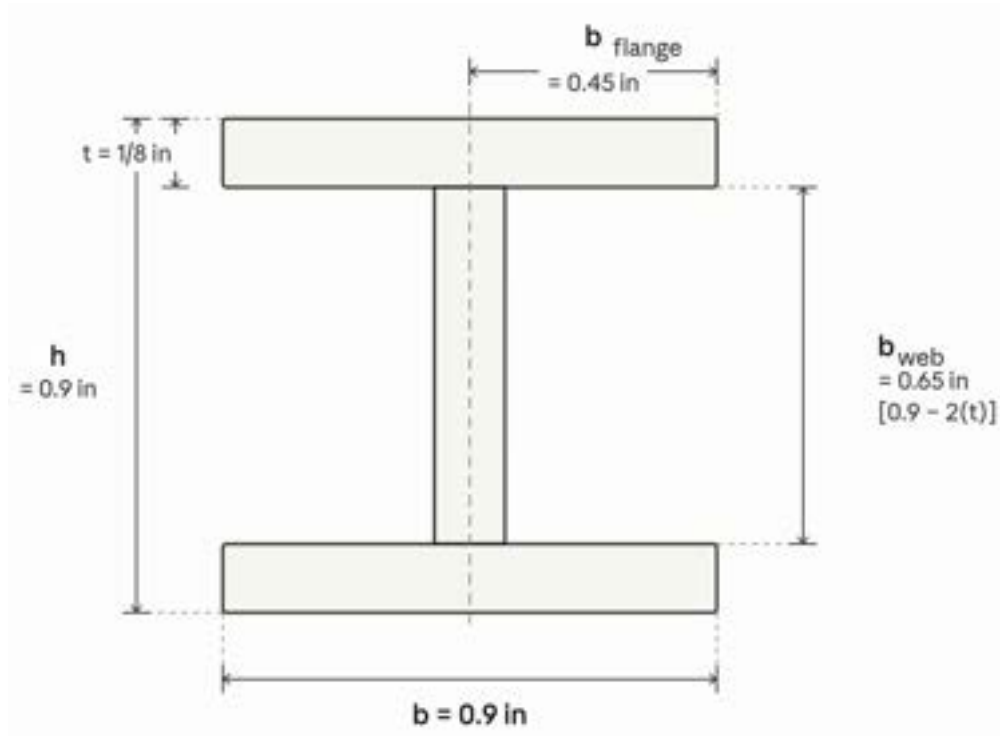


Figure 12: I-beam cross section with $b_{\text{flange}} = 0.45 \text{ in}$ and $b_{\text{web}} = 0.65 \text{ in}$.

The critical stress for local plate buckling is:

$$\sigma_{\text{plate}} = \frac{k\pi^2 E}{12(1 - \nu^2)} \left(\frac{t}{b} \right)^2 \quad (20)$$

Simply supported boundary conditions are assumed everywhere (on all sides) for the web, so

$K_{web} = 4$. On the other hand, the half-flange plate being analyzed has one free side, along the length of the spar, and the rest are assumed simply supported, so $K_{flange} = 0.43$. The geometry parameters are:

$$K_{web} = 4, \quad K_{flange} = 0.43, \quad t = \frac{1}{8} \text{ in}, \quad \nu = 0.009$$

With the value for ν retrieved from the MatWeb material properties for American balsa wood.

$$\sigma_{cr,web} = \frac{k\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b_{web}} \right)^2 = 413.700 \text{ MPa} \quad (21)$$

$$\sigma_{cr,flange} = \frac{k\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b_{flange}} \right)^2 = 92.789 \text{ MPa} \quad (22)$$

The critical stresses above for both the plate buckling of our web and half the flange are significantly larger than the maximum bending stress imparted on the wing during takeoff with the 1.5 safety factor: $\sigma_z = 5.0812 \text{ MPa}$. Therefore, we conclude that no plate buckling will occur along the web or flange during the takeoff flight condition.

4.1.3 Part 3: Idealization of Wing Cross Section

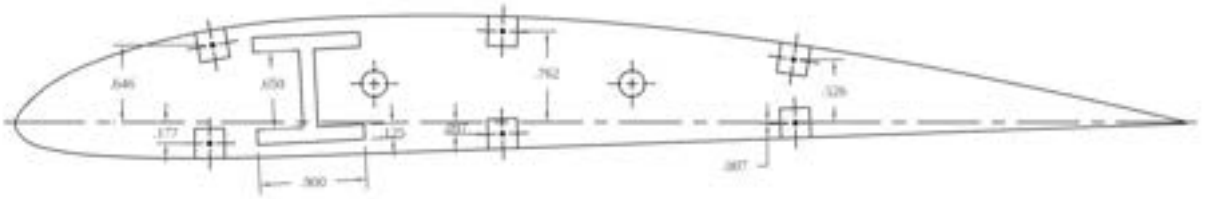


Figure 13: Cross Section showing Stringer Dimensions

The stringer distances from the neutral axis are:

$$y_1 = 0.646 \text{ in}, \quad y_2 = 0.177 \text{ in}, \quad y_3 = 0.762 \text{ in}, \quad y_4 = 0.097 \text{ in}, \quad y_5 = 0.526 \text{ in}, \quad y_6 = 0.007 \text{ in}$$

Each stringer has a square cross-section of $0.25 \times 0.25 \text{ in}$, so the boom area and second moment of inertia of each stringer are as follows:

$$B_{st} = 0.25^2 = 0.0625 \text{ in}^2, \quad I_{xx} = I_{yy} = \frac{0.25^4}{12} \text{ in}^4$$

The total, idealized second moment of area of the wing box cross-section about the x -axis is:

$$\begin{aligned} I_{xx,\text{total}} &= I_{xx,\text{beam}} + I_{xx,\text{stringers}} \\ &= I_{xx,\text{beam}} + B_{st} [y_1^2 + y_2^2 + y_3^2 + y_4^2 + y_5^2 + y_6^2] \\ &= 0.1198 \text{ in}^4 \end{aligned} \quad (23)$$

4.1.4 Part 4: Stiffener Buckling of Stringers

The most critical stringer is stringer 3 at $y_{\text{max}} = 0.762 \text{ in}$, with $M_{\text{max}} = -6.845 \text{ N}\cdot\text{m}$. We are assuming pinned-pinned boundary conditions for the stringers for the same reasoning as explained for the spar with $E = 3.4 \text{ GPa}$ and $A = B_{st} = 0.25^2 \text{ in}^2$.

$$\sigma_{cr} = \frac{M_{\text{max}} y_{\text{max}}}{I_{xx,\text{total}}} = 2.6697 \text{ MPa} \quad \rightarrow \quad l_e = \left[\frac{\pi^2 E I_{yy}}{\sigma_{cr} \cdot A} \right]^{1/2} = 0.2056 \text{ m} \quad (24)$$

Since $l_{e,\text{stringer}} < l_{e,\text{beam}}$, the rib spacing must be recalculated using the stringer effective length:

$$N_{\text{ribs}} = \frac{b/2}{l_{e,\text{stringer}}} = \frac{0.475}{0.2056} = 2.31 \quad \rightarrow \quad N_{\text{ribs}} = 3 \quad \rightarrow \quad \text{Spacing} = \frac{b/2}{3} = 0.1583 \text{ m} \quad (25)$$

Therefore, rounding up from the calculated number of ribs to three ribs gives $N_{\text{ribs}} = 4$, since one rib is needed at each end of the wing; therefore, our rib configuration will consist of a rib on each end of the wing and two ribs evenly spaced in the middle, yielding a rib spacing of 0.1583 m.

4.2 Direct and Shear Stress Distribution Analysis

4.2.1 Direct Stress

The direct stress distribution of the I-beam is given by the following equation:

$$\sigma_z = \frac{M_x (I_{yy}y - I_{xy}x)}{I_{xx}I_{yy} - I_{xy}^2} + \frac{M_y (I_{xx}x - I_{xy}y)}{I_{xx}I_{yy} - I_{xy}^2} \quad (26)$$

Since bending is only about the x -axis, $M_y = 0$. Also, as stated previously, we are assuming $I_{xy} = 0$ since our value for I_{xy} was negligible, so our I-beam is nearly doubly-symmetric which simplifies the expression to:

$$\sigma_z = \frac{M_x y}{I_{xx}} \quad (27)$$

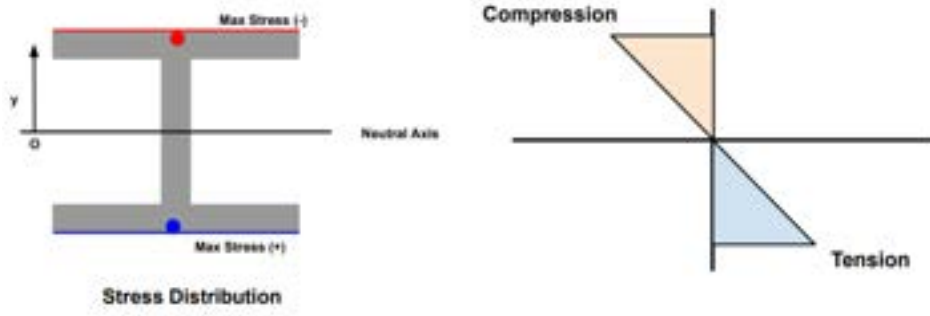


Figure 14: Spar Cross Section Direct Stress Distribution

Figure 14 above shows the direct stress distribution of our spar, and it can be seen that the top and bottom of the I beam experience the maximum compressive and tensile stress, respectively. From our spar buckling analysis, we found the maximum direct stress at $y_{max} = 0.45$ in:

$$\sigma_{max} = \frac{|M_{max}| y_{max}}{I_{xx, spar}} = 5.0812 \text{ MPa} \quad (28)$$

4.2.2 Shear Stress

The shear flow distribution of the I-beam is given by the following equation:

$$q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s tx \, ds - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s ty \, ds + q_{s,0} \quad (29)$$

Since we are assuming the only applied shear force during takeoff is the takeoff load, T , $S_x = 0$. Also, our I-beam is nearly doubly-symmetric, so we are assuming $I_{xy} = 0$, and $q_{s,0} = 0$ since the spar is an open section, which simplifies the expression to:

$$q_s = - \frac{S_y}{I_{xx}} \int_0^s ty \, ds \quad (30)$$

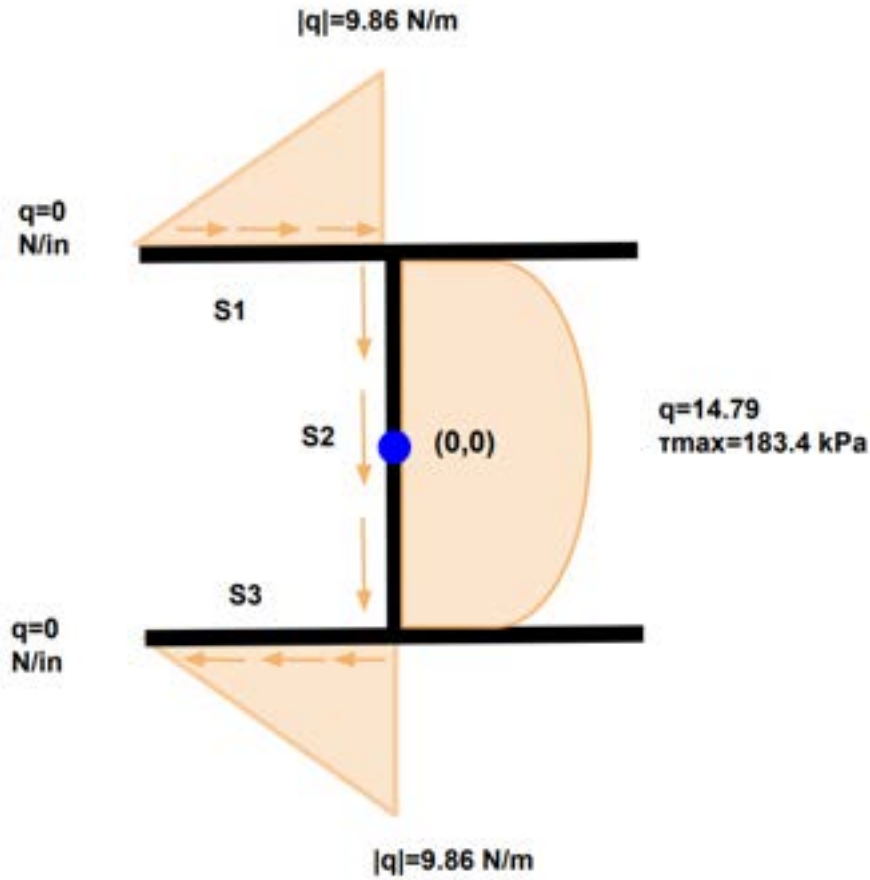


Figure 15: Spar Cross Section Shear Distribution

Figure 15 above shows the shear flow through one side of the I-beam, as well as the magnitude and location of the maximum shear stress. The yellow arrows indicate the shear flow on the left side of the I-beam, and the shaded regions indicate the corresponding magnitude of the shear flow along those portions of the I-beam. Since the I-beam is doubly symmetric, the shear flow is also symmetric across the I-beam, therefore, we calculated the shear flow for one side. The integration paths s_1 , s_2 , and s_3 are defined along the top flange, web, and bottom flange respectively, starting at the top-left free edge.

Along s_1 , $y = 0.45$ in:

$$q_{12} = -\frac{S_y t}{I_{xx}} \int_0^{s_1} (0.45 \text{ in}) ds = -\frac{S_y t}{I_{xx}} (0.45 s_1) \quad (31)$$

$$q_2 = q_{12} \Big|_{s_1=0.45 \text{ in}} = -\frac{S_y t}{I_{xx}} (0.45 \text{ in})^2 \quad (32)$$

Along s_2 , $y = 0.45 - s_2$:

$$q_{23} = -\frac{S_y t}{I_{xx}} \int_0^{s_2} (0.45 - s_2) ds + q_2 = -\frac{S_y t}{I_{xx}} \left[0.45 s_2 - \frac{1}{2} s_2^2 \right] + \left[-\frac{S_y t}{I_{xx}} (0.45 \text{ in})^2 \right] \quad (33)$$

$$q_{23} = \frac{S_y t}{I_{xx}} \left[\frac{1}{2} s_2^2 - (0.45 \text{ in}) s_2 - (0.45 \text{ in})^2 \right] \quad (34)$$

$$q_3 = q_{23} \Big|_{s_2=0.9 \text{ in}} = -\frac{S_y t}{I_{xx}} (0.45 \text{ in})^2 \quad (35)$$

Along s_3 , $y = -0.45 \text{ in}$:

$$q_{34} = -\frac{S_y t}{I_{xx}} \int_0^{s_3} (-0.45) ds + q_3 = \frac{S_y t}{I_{xx}} (0.45 \text{ in}) s_3 - \frac{S_y t}{I_{xx}} (0.45 \text{ in})^2 \quad (36)$$

$$q_4 = q_{34} \Big|_{s_3=0.45 \text{ in}} = \frac{S_y t}{I_{xx}} (0.45 \text{ in})^2 - \frac{S_y t}{I_{xx}} (0.45 \text{ in})^2 = 0 \quad (37)$$

The shear flow returns to zero at the free edge, confirming the open-section boundary condition is satisfied. Together, these equations and the values of the shear flow at the end of the flange and connection points between the flange and the web were used to create Figure 15 above.

Maximum Shear Stress

The maximum shear stress occurs at the neutral axis, evaluated at $s_2 = 0.45 \text{ in}$ and using the unit conversion in Appendix Section A.1:

$$q_{23} \Big|_{s_2=0.45 \text{ in}} = \frac{S_y t}{I_{xx}} \left[\frac{1}{2} s_2^2 - (0.45 \text{ in}) s_2 - (0.45 \text{ in})^2 \right] = -14.789 \text{ N/in} \quad (38)$$

$$\tau_{\max} = \frac{|q|}{t} = \frac{14.789 \text{ N/in}}{0.125 \text{ in}} = 118.31 \text{ N/in}^2 = 183,400 \text{ N/m}^2 = 183.4 \text{ kPa} \quad (39)$$

4.3 Failure Analysis

The principal stresses are calculated using the stress state $\sigma_x = \sigma_z$, $\sigma_y = 0$, with material properties $\sigma_{UT} = 14 \text{ MPa}$ and $\sigma_{UC} = 7 \text{ MPa}$, and the previously derived $\sigma_{z,\max} = 5.08 \text{ MPa}$ and $\tau_{\max} = 183.4 \text{ kPa}$.

The major and minor principal stresses are:

$$\sigma_I = \frac{\sigma_{z,\max}}{2} + \frac{1}{2} \sqrt{\sigma_{z,\max}^2 + 4\tau_{\max}^2} = 5.087 \text{ MPa} \quad (40)$$

$$\sigma_{II} = \frac{\sigma_{z,\max}}{2} - \frac{1}{2} \sqrt{\sigma_{z,\max}^2 + 4\tau_{\max}^2} = -6.613 \text{ kPa} \quad (41)$$

4.3.1 Coulomb–Mohr Failure Criterion

Applying the Coulomb–Mohr criterion for a brittle material with unequal tensile and compressive strengths:

$$\frac{\sigma_I}{\sigma_{UC}} + \frac{\sigma_{II}}{\sigma_{UT}} = \frac{5.087}{7} + \frac{6.614 \times 10^{-3}}{14} = 0.727 \quad (42)$$

Since $0.7 < 0.727 < 0.9$, the component is within the acceptable safety range of the design criterion, based on the maximum direct and shear stresses calculated with a 1.5 safety factor applied to the takeoff load, T.

5 Wing Critical Design (DK)

A CAD (Computer-Aided Design) model of the prototype was created using Onshape®. An engineering drawing with front, back, and top views is available in the Appendix (A.2). The four ribs (one root rib, one tip rib, and two mid ribs), one I-beam spar, and six square-cross-section stringers were constructed separately and assembled in Assembly mode. The mate feature (primarily plane mates with fixed offsets for each dimension) was used to connect the various components, representing the role of epoxy in the prototype. Three 5.5 mm clearance holes were placed in the root rib for the three M5 screws used to mount the wing and two 5.5 mm clearance holes were placed in the tip rib to attach weights to the wing. The Monokote skin and 1/32" balsawood leading and trailing edge skins were not considered in the analysis as these components served a primarily aesthetic purpose and provided the wing a smoother shape.

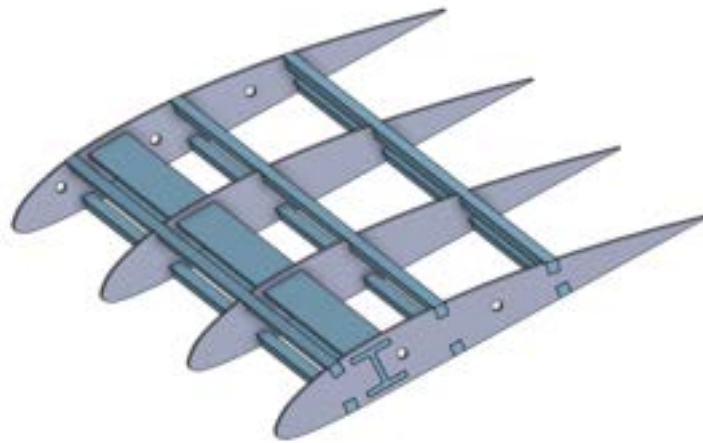


Figure 16: CAD Model

The assembled model was then transferred as a STEP (format ISO 10303) file into Ansys Inc.® for finite element analysis (FEA). A mesh of linear elements of size 0.01 m was used in the analysis to balance adequate accuracy with reasonable solving time. A "Fixed Support" constraint was placed at the three screw holes of the root rib and an upward "Constant Force" was applied along the surface of the tip rib.

Structural failure was determined by three criteria: equivalent (direct) stress, shear stress and total deformation as shown in the three figures below. We expected the model to fail at the minimum at load at which either of the stresses exceeded 7 MPa (ultimate compressive stress) or the total deformation exceeded 1.2%. Due to the variation in stress and deformation at each point, it was also important to note the location at which this minimum load caused the model to fail.

Initial FEA (Prior to Testing)

In the initial analysis, the entire surface of the root rib was selected for the fixed support constraint. This yielded a breaking stress of 18.0 N (1.83 kg) which is 1.25 times the limit load. Hence, we expected that the prototype would fail along the I-beam at this load as seen in the figure below.

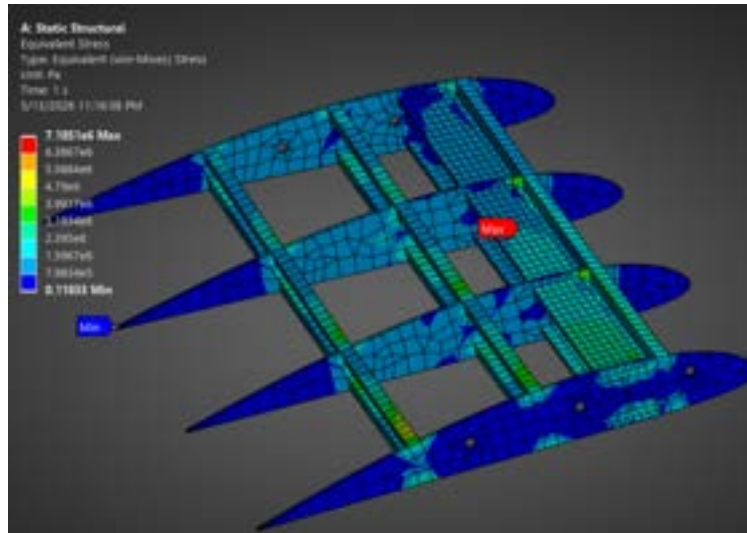


Figure 17: Initial Analysis - Equivalent Stress

This model was unsuccessful at predicting when and where the prototype failed. The model was revised to better capture how the wing was mounted to the fixed support (i.e. using the M5 screws).

Revised FEA (Following Testing)

Following testing, it was determined that only the three M5 screw holes should be selected as fixed supports. This significantly impacted our stress and deformation values. The maximum direct stress of 7 MPa was reached when a force of 1.8 N (184 grams) was applied to the tip rib. This is one-eighth of the expected limit load (14.4 N or 1.47 kg). As seen in the figure below, the three outer ribs, the spar and the stringers experienced little to no direct stress (< 1 Pa). Unfortunately, the regions around the three M5 screw holes of the root rib, particularly the central one, experienced sufficient stress to cause failure. From this, it is clear that the wing will fail prematurely due to the weak root rib, which is consistent with what we saw during the testing phase.

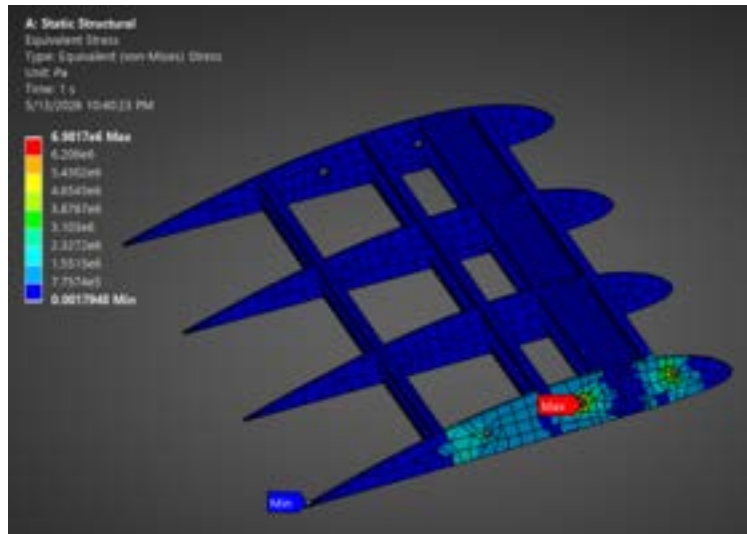


Figure 18: Equivalent Stress Analysis

The shear stress distribution across the model showed significantly less variation with maximum values of 2.30 MPa in the positive direction and 2.33 MPa in the negative direction, both of which occurred in the root rib. While this may not have been sufficient to cause the model to fail, it would have caused significant strain on it. The remaining components of the model experienced roughly a constant shear stress of 0.25 MPa throughout.

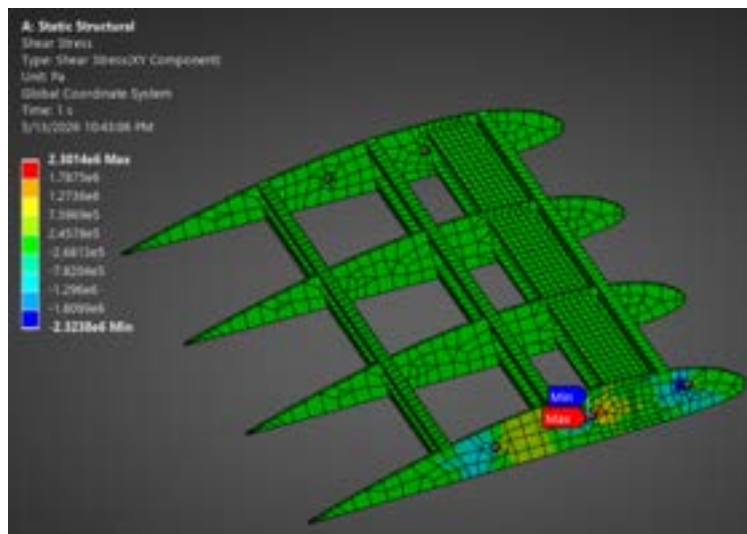


Figure 19: Shear Stress Analysis

The maximum deformation predicted by the software was 0.3%, significantly below the strain needed to cause failure (1.2%). The deformation increased with distance from the root rib. The greatest deformation was observed at the trailing edge portion of the tip rib and the least was observed at the surface of the root rib.

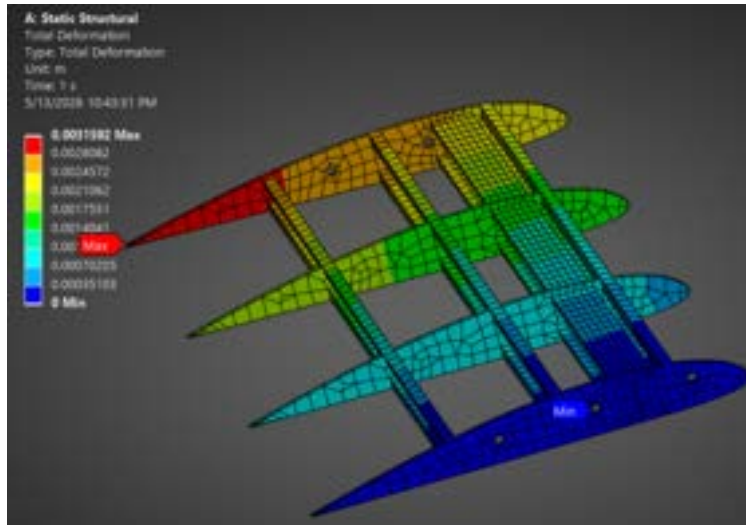


Figure 20: Total Deformation Analysis

Therefore, the actual failure of the prototype should have occurred at a 1.8 N load along the root rib, specifically around the edges of the screw holes.

6 Wing Prototype (EI)

The final prototype was built as a semi-span wing based on the geometry developed in the conceptual and preliminary design sections. The final design used a full span of $b = 0.95$ m, so the fabricated semi-span length was approximately 0.475 m. The chord was $c = 0.25$ m, giving a plan-form area of $S = 0.2375$ m², a lift coefficient of $C_l \approx 0.45$ and an angle attack of approximately 3 ° consistent with the final aerodynamic design values. The completed prototype had a measured mass of approximately 83 g.

The internal structure consisted of a balsa wood I-beam spar, ribs, six stringers, leading and trailing edge balsa sheets, epoxy joints, and a Monokote covering. The I-beam spar served as the primary bending member, while the ribs maintained the Clark-Y airfoil shape and transferred load between the wing surface and the spar. The six stringers were arranged with three on the upper surface and three on the lower surface, consistent with the preliminary design layout. Based on the stringer buckling requirement, the final design used four ribs total across the semi-span, giving a rib spacing of approximately 0.1583 m.

Fabrication began by cutting the Clark-Y rib profiles and preparing the balsa wood spar, stringers, and edge support pieces. The I-beam spar was built from 1/8 in balsa wood, with the web and flanges bonded together using epoxy. The spar was inserted into the rib slots and aligned along the span. The stringers were then fitted into their corresponding slots and bonded to the ribs and spar using epoxy with three stringers on the upper surface and three on the lower surface to support the wing shape and reduce the unsupported length of the structure. The joints between the ribs, spar, and stringers were bonded using epoxy and clamped during curing to keep the structure straight.

Once the internal frame was assembled, balsa sheet pieces were added near the leading and trailing edges to improve local stiffness and provide smoother surface transitions. Because the balsa sheet was initially too stiff to bend cleanly around the airfoil profile, water was applied to soften the wood fibers. The softened balsa was gradually bent around the leading edge to match the Clark-Y shape and then bonded in place with epoxy. Clamps were also used during curing to hold the sheet in place. The trailing-edge sheet was positioned approximately 3 in from the lower short side and approximately 5 in from the upper long side, helping define the aft surface transition and complete the wing profile. After the internal frame and edge surfaces were complete, the wing was covered with Monokote. The Monokote was heat-shrunk over the frame to create a smooth aerodynamic surface.

The physical prototype generally followed the CAD design, including the rib locations, spar placement, stringer arrangement and overall semi-span geometry. The main difference between the CAD model and the fabricated prototype was the leading edge construction which was done to provide structural strength to the prototype.

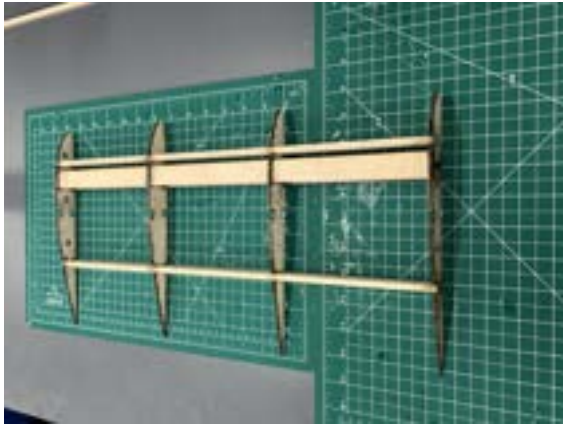


Figure 21: Partly Assembled Wing



Figure 22: Wing with the trailing and leading edge



Figure 23: Clamping of the wing when adding the trailing and leading edge



Figure 24: Side view of the wing



Figure 25: Midway through wrapping the wing with monokote



Figure 26: Fully wrapped the wing with the monokote

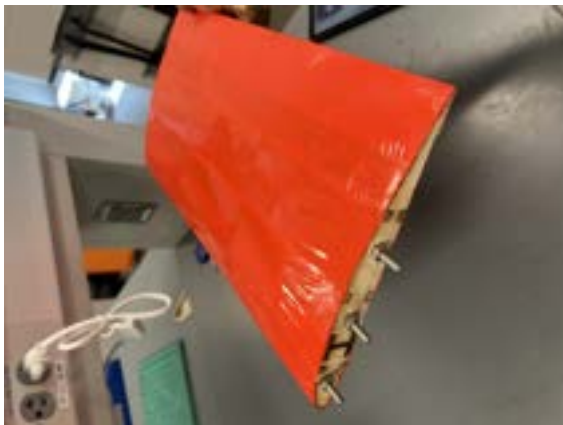


Figure 27: Fully assembled wing

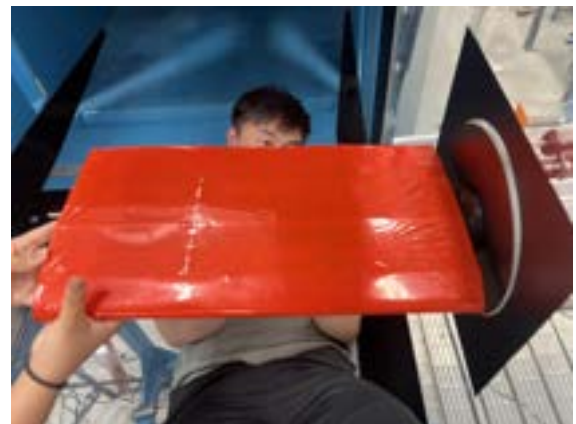


Figure 28: Mounting the Wing in the Wind Tunnel for testing

Table 2: Bill of Materials

Component	Material / Size	Quantity Purchased	Unit Price	Total Cost
Stringers	Balsa wood square rods, 1/4 in $\times$ 1/4 in, 3 ft length	6 rods	\$0.50/rod	\$3.00
I-beam spar material	1/8 in balsa wood strips/sheets	2 strips	\$3.00/strip	\$6.00
Ribs	1/8 in balsa wood sheets, 12 in $\times$ 6 in	2 sheets	\$2.00/sheet	\$4.00
Leading/trailing edge flat pieces	1/32 in balsa wood strips	2 strips	\$3.00/strip	\$6.00
Adhesive	JB Weld epoxy	4 units	\$3.00/unit	\$12.00
Covering	Monokote roll, 12 in $\times$ 20 in	2 rolls	\$5.00/roll	\$10.00
Total				\$41.00

7 Design Reflection (EI) (YB) (TH) (DK)

There were some issues in the testing of our wing. On our first iteration our wing was unable to withstand the limit load and broke the rib which the wing was fixed upon. It is worth noting that at this time, there was no other failure. We tried again on the other side of our beam using the remaining outer rib and encountered a similar issue. In the end, the wing had to be clamped down to the table. Overall our wing was able to withstand a load of 1.3 kg, which was not our predicted 2 kg ($1.4 \times \text{limit load}$) or 1.8 kg ($1.25 \times \text{limit load}$).

Reflecting on our design process, we revised our FEA model to better account for how the wing was mounted using the three M5 screws of the root rib. Following this revision, the failure load was recomputed and predicted to be 0.18 kg, approximately one-tenth of the previously predicted load. This is likely why the model initially failed at the root rib when a 0.5 kg weight was attached. We were able to continue adding weights following the change in clamping procedure. The discrepancy between the actual weight our wing held before breaking and our predicted values was attributed to the clamping method we used.

Elements of the conceptual design, preliminary design, and critical design process were all critical to ensuring the success of our design. In the conceptual design, our analysis allowed us to calculate and decide values for our span, chord, and angle of attack that could withstand and operate in the two loading conditions. Also, this section enabled us to determine that the takeoff condition was the critical flight condition, which heavily impacted our preliminary analysis. The preliminary analysis was critical to determining the number of ribs and rib spacing, as well as ensuring the spar would be able to withstand the maximum shear and direct stresses. The buckling analysis of the stringers revealed that we needed four ribs instead of three. Lastly, the critical design process was essential because it helped us realize that the distribution of shear stress, direct stress and deformation were significantly altered by which surfaces the fixed support condition was applied to. This in turn reduced the breaking load by a factor of 10 and changed the location at which failure first occurred.

Overall, the design process demonstrated that the main wing structure performed reasonably well. The I-beam spar, ribs, and stringers effectively carried the applied load and maintained the wing's shape, and the stringer buckling analysis proved especially important in determining the final rib spacing. However, testing revealed that the root connection was the weakest part of the design, and while we successfully predicted that the wing would fail due to bending of the I-beam near the root, the actual failure behavior and final weight deviated somewhat from our expected values.

Several improvements could be made to both the analysis and design in a future iteration. On the modeling side, we initially applied the fixed boundary condition across the entire root rib surface, which did not accurately reflect how the wing was mounted during testing. A more realistic approach would be to constrain only the specific bolt hole locations in FEA, as this would better capture the true stress distribution, more faithfully replicate the actual failure mode, and

allow potential issues to be identified and addressed earlier in the design process. On the design side, we should have made the root rib out of basswood instead of balsa wood and increased its thickness from 1/8 in to 1/4 in.

8 References

1. “Balsa Wood Material Properties.” *Make It From*,
www.makeitfrom.com/material-properties/Balsa. Accessed 13 May 2026.
2. “MatWeb: Material Property Data.” *MatWeb*,
www.matweb.com/index.aspx. Accessed 13 May 2026.

A Appendices

A.1 Unit Conversions (TH)

All calculations were performed in SI units. The following conversions were applied throughout.

Length

$$1 \text{ in} = 0.0254 \text{ m}$$

Quantity	Given	SI Value
$y_{\max}$ (beam)	0.45 in	$0.45 \times 0.0254 = 0.01143 \text{ m}$
b_{flange}	0.45 in	$0.45 \times 0.0254 = 0.01143 \text{ m}$
b_{web}	0.65 in	$0.65 \times 0.0254 = 0.01651 \text{ m}$
t	0.125 in	$0.125 \times 0.0254 = 0.003175 \text{ m}$
$y_{\max}$ (stringer)	0.768 in	$0.768 \times 0.0254 = 0.0195072 \text{ m}$

Area

$$1 \text{ in}^2 = (0.0254 \text{ m})^2 = 6.4516 \times 10^{-4} \text{ m}^2$$

Quantity	Given	SI Value
A (I-beam)	0.306 in^2	$0.306 \times 6.4516 \times 10^{-4} = 1.9742 \times 10^{-4} \text{ m}^2$
A (stringer)	0.0625 in^2	$0.0625 \times 6.4516 \times 10^{-4} = 4.0323 \times 10^{-5} \text{ m}^2$
$\tau_{\max}$ (I-beam)	118.31 N/in^2	$118.31 \div 6.4516 \times 10^{-4} = 183,400 \text{ N/m}^2$

Second Moment of Area

$$1 \text{ in}^4 = (0.0254 \text{ m})^4 = 4.162314256 \times 10^{-7} \text{ m}^4$$

Quantity	Given	SI Value
I_{xx} (I-beam)	0.037 in^4	$0.037 \times 4.1623 \times 10^{-7} = 1.5401 \times 10^{-8} \text{ m}^4$
I_{yy} (I-beam)	0.015 in^4	$0.015 \times 4.1623 \times 10^{-7} = 6.2435 \times 10^{-9} \text{ m}^4$
$I_{xx,\text{total}}$	0.1198 in^4	$0.1198 \times 4.1623 \times 10^{-7} = 4.9885 \times 10^{-8} \text{ m}^4$
I_{yy} (stringer)	$0.25^4/12 \text{ in}^4$	$3.2552 \times 10^{-4} \times 4.1623 \times 10^{-7} = 1.3548 \times 10^{-10} \text{ m}^4$

Force and Moment

$$1 \text{ N}\cdot\text{m} = 1 \text{ kg}\cdot\text{m}^2\cdot\text{s}^{-2}$$

Quantity	Expression	Value
Tip load P	1.5×9.6075	14.411 N
Weight W	1.5×9.81	14.715 N
$M_{\max}$	$1.5 \times 9.6075 \times 0.475$	6.845 N·m

A.2 Engineering Drawings (DK)

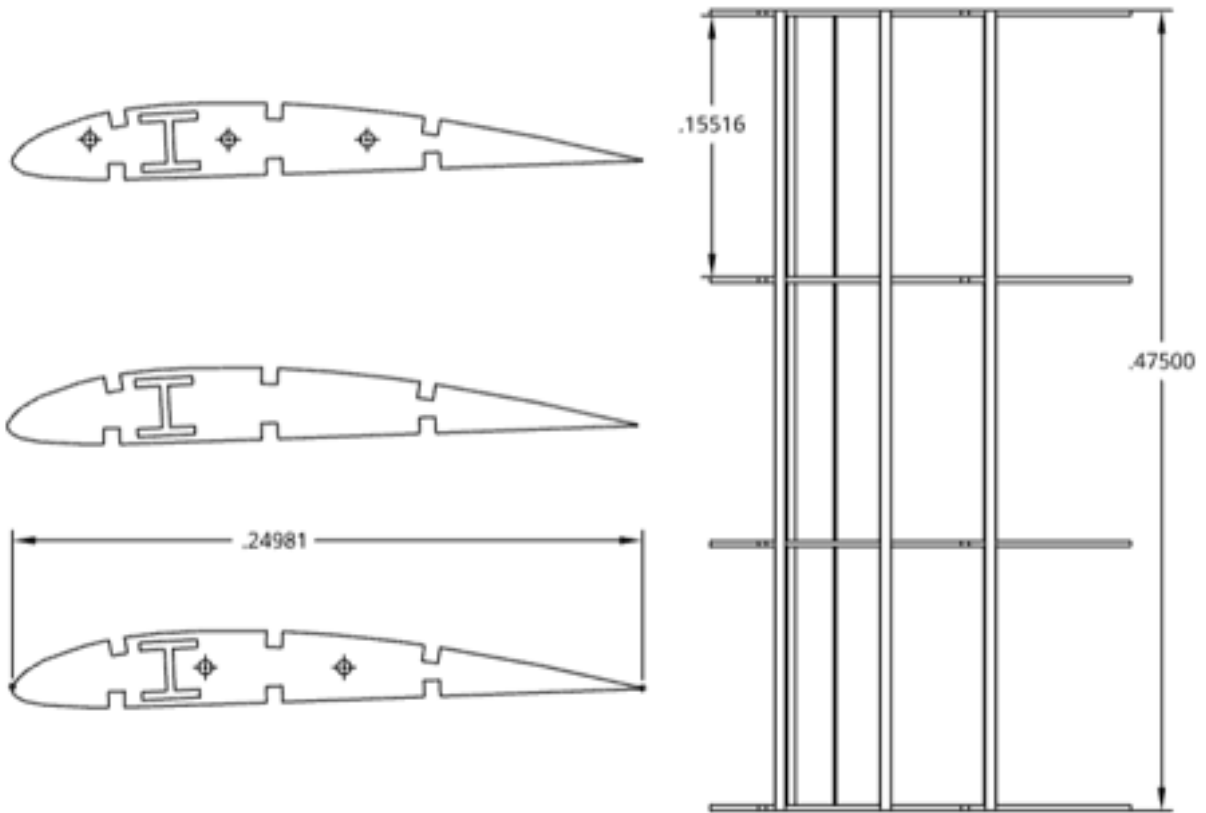


Figure 29: Ribs Views and Top View of Model